

tring and easyclimate: two R packages to analyze long-term responses of forest ecosystems

PI channel
shopping

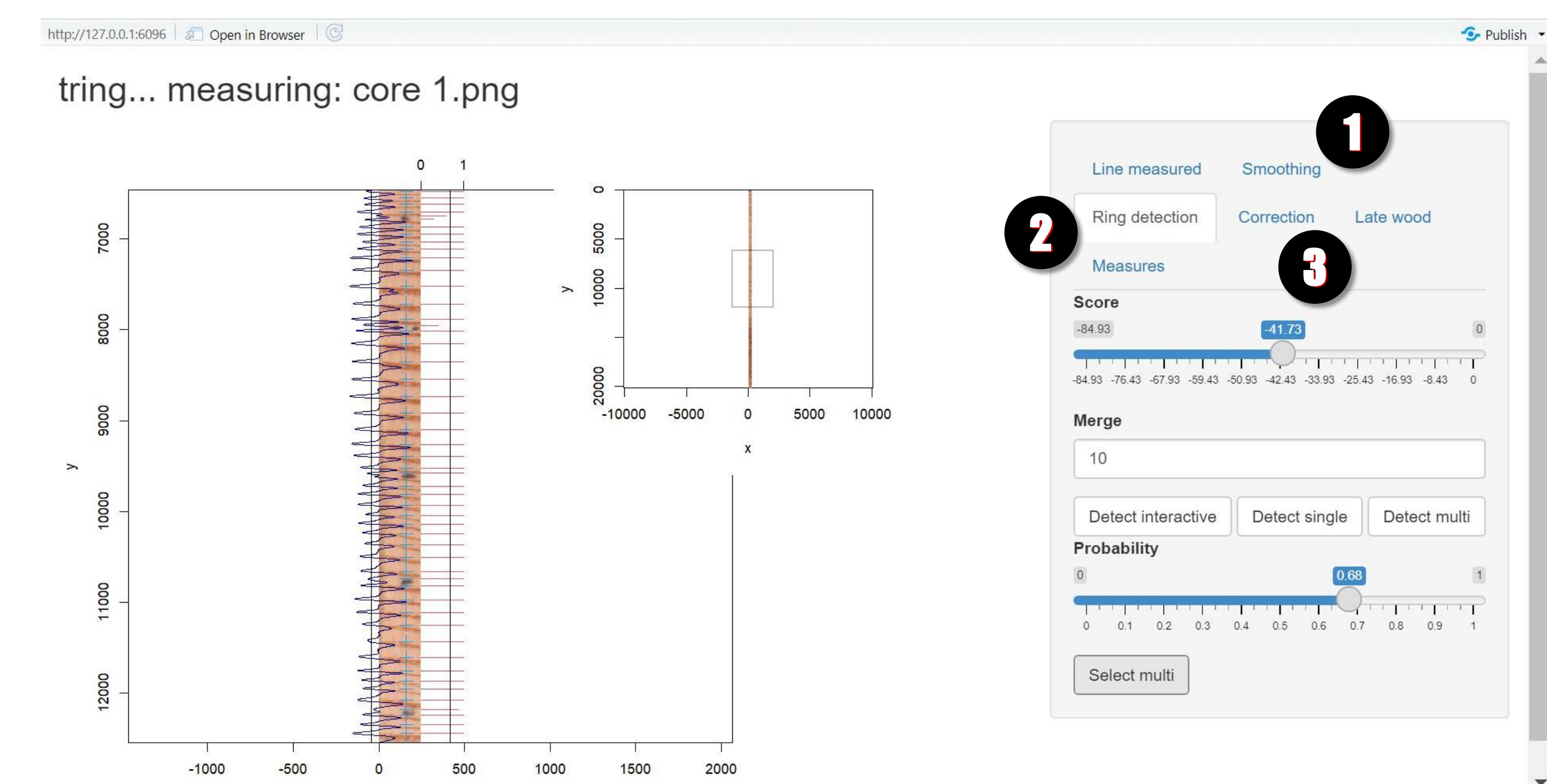
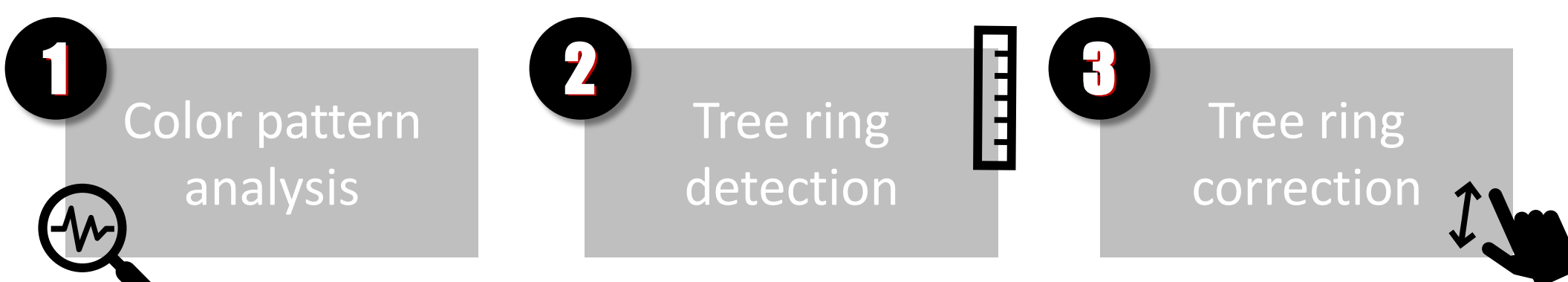
Are you tired of dating thousands of tree cores without even detecting the missing rings?

tring is the final R package that you were waiting for!!

tring package facilitates tree ring width measurement on scanned images of wood cores.

How does this magic work?

3 basic steps...



...with 3 possible methodologies

depending on the bands used for tree ring detection along the wood core

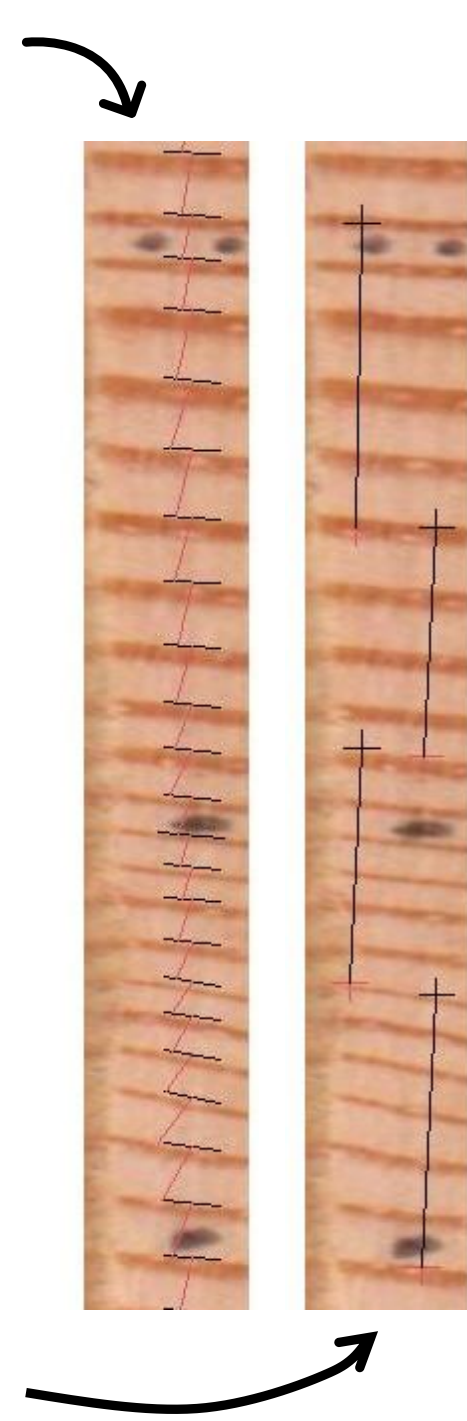
Single

Multiple

Interactive

With **multiple band detection**, the package calculates the probability of existence of each ring based on the color pattern analysis over all bands. Once the user selects the probability threshold to detect a ring, the package calculates the ring slope according to all bands and performs orthogonal ring width measurements

With the **interactive band detection**, the user draws the bands for ring width measurements and they can be manually inclined for considering tree ring angle



User- friendly interface with:

- zoom in/ zoom out
- navigation through the image
- comparison of images
- detection of discontinuities in the cores

We are working on tring development right now. The first release will be ready very soon!
In the future we will incorporate late/ early wood differentiation and the blue intensity method.

"tring has simply changed my life"



Dr. Calatayud



Dr. Cruz-Alonso

Is climatic information very hard to get unless you measure it yourself?

AS SEEN ON
GitHub

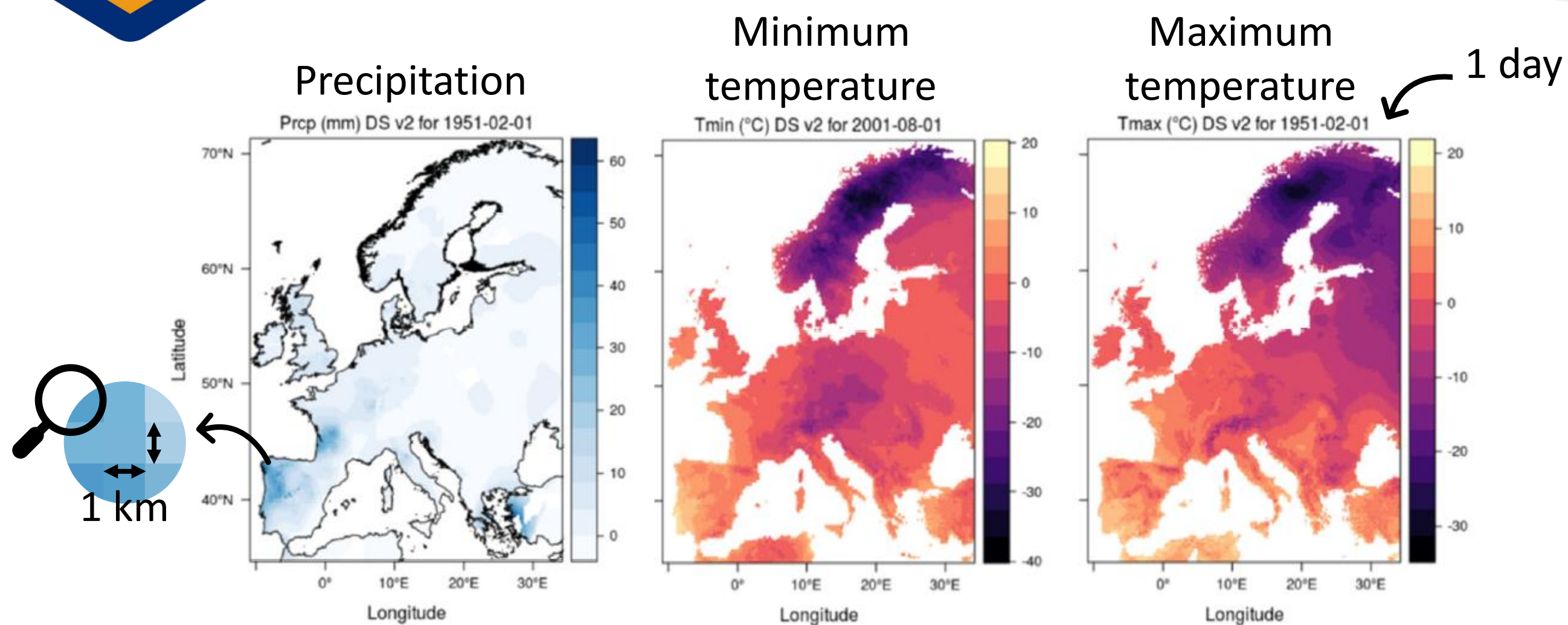
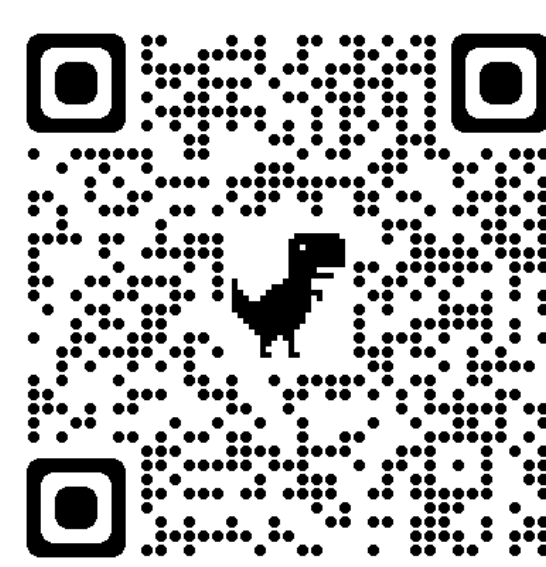
TeleClimate,
how may I
help you?

Download easyclimate R package NOW and enjoy the advantages!!



easyclimate package facilitates the downloading and processing of high-resolution (1 km²) daily climate data for Europe

We are serious! You can actually download and try easyclimate package in <https://github.com/VeruGHub/easyclimate/>



The climatic database was first published by Moreno & Hasenauer in 2015 (10.1002/joc.4436) and improved and updated in 2018 by Rammer et al. Currently the database is being updated again to include information from 1950 to 2020 for all Europe

get_daily_climate

Is the main function in easyclimate. It extracts daily climate data for a given set of points or polygons

```
get_daily_climate(  
  coords = NULL,  
  climatic_var = "Prcp",  
  period = NULL,  
  output = "df"  
)
```

Coordinates (longitude and latitude) as points or polygons

"Prcp", "Tmin" or "Tmax"

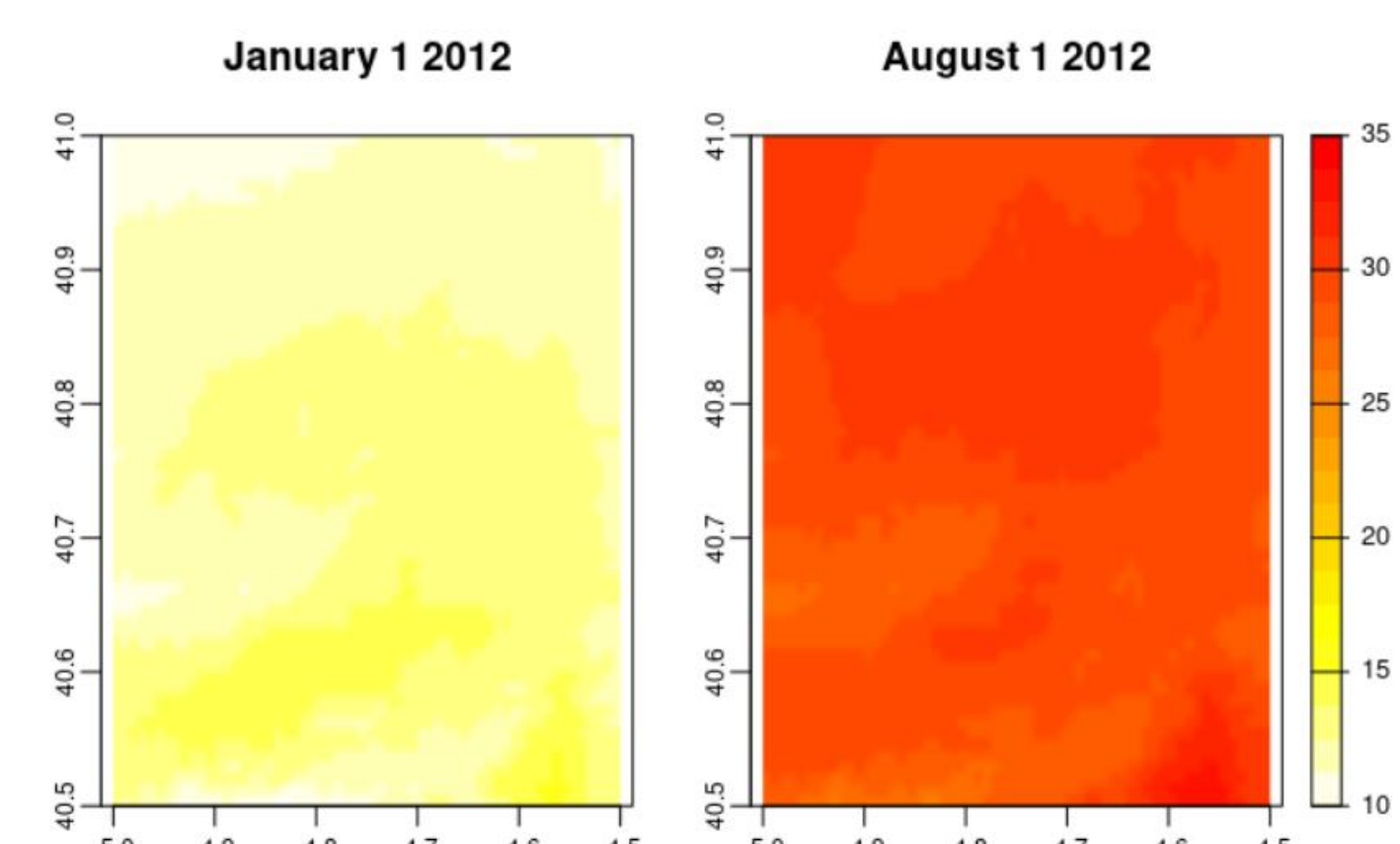
A year (e.g. 1988), a date (e.g. 1988-12-26) or a period of years or dates

"df" returns a data frame with daily climatic values for each point/polygon, and "raster" returns a terra::SpatRaster() object

The result with output = "df" looks like this:

ID_coords	x	y	date	Prcp
1	-5.36	37.4	2001-01-01	945
1	-5.36	37.4	2001-01-02	12
1	-5.36	37.4	2001-01-03	205

...and with output = "raster", like this:



"Since I started using easyclimate I enjoy every night of long-format data sets with my favorite climatic variables"

Download easyclimate NOW and you will get FREE training for tring where you will measure 500 tree cores for us... don't miss the opportunity!

easyclimate: veronica.cral@gmail.com
tring: julenastigarraga@gmail.com
j.calatayud.ortega@gmail.com

Contact us if you have any question or suggestion

easyclimate: @Veruitter
tring: @J_astigarraga

easyclimate: VeruGHub
tring: Julenasti · Joacala

References:

Cruz-Alonso V¹, Rodríguez-Sánchez F², Pucher C³, Ratcliffe S⁴, Astigarraga J⁵, Neumann M³, Ruiz-Benito P⁵ (2021). easyclimate: Easy access to high-resolution daily climate data for Europe. <https://verughub.github.io/easyclimate/>

Astigarraga J⁵, Cruz-Alonso V¹, Andivia E⁶, Ballesteros-Cánovas JA⁷, Herrero A⁸, Madrigal J⁹, Granda E⁵, Silla F¹⁰, Calatayud J¹¹ (soon...). tring: tree ring measurement